

Phase II Laboratory Report

*Construction of an Atlas of L-functions and
Systematic Search for Exceptional Symmetry*

Toward the structure of the Langlands group L_Q

Phase	II — Atlas construction & exceptional-symmetry search
Atlas size	1 200 synthetic L-function objects (250 CM + 550 GEN + 400 MOD)
Enriched dimension	30 (original 22 arithmetic fingerprints + 8 L-function statistics)
Graph	k-NN similarity graph, k = 15, 13 716 undirected edges
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Abstract

Building directly on the Phase-I statistical fingerprint laboratory, we construct a synthetic atlas of 1 200 L-function objects, enrich each object with additional local and global statistics (low-lying zeros, root numbers, analytic conductors, pair-correlation proxies), and convert the resulting 30-dimensional feature space into a weighted k-nearest-neighbour similarity graph. Spectral analysis of the normalised graph Laplacian recovers a clean community structure that separates the CM locus from the non-CM locus and further partitions the modular/generic cloud. An anomaly score identifies the most exceptional objects; at the present synthetic resolution they lie predominantly inside the more variable CM cloud. A rudimentary tensor-product test already produces synthetic tensors whose nearest-neighbour distances fall inside the existing atlas, furnishing a first computational probe of the multiplicative structure predicted by Langlands. The apparatus is now ready for the substitution of genuine LMFDB data and for a systematic hunt for sporadic or exceptional symmetry.

1. Goals of Phase II

Phase I established that a uniform 22-dimensional fingerprint built from local Frobenius data already recovers the CM / non-CM dichotomy and the modularity identification of weight-2 newforms with elliptic curves. Phase II advances the experimental programme in four concrete directions:

- Enrichment of every object with synthetic analogues of the global invariants of genuine L-functions (zeros, root numbers, conductors).
- Construction of a similarity graph and extraction of its spectral embedding and community structure.
- Systematic search for exceptional / outlier objects that could foreshadow new structure inside L_Q .
- First computational test of tensor-product compatibility — the multiplicative structure that Langlands reciprocity demands.

2. Enrichment of the Fingerprint

Eight new coordinates were appended to the original 22-dimensional vector:

New coordinate	Description
mean_gap, var_gap, min_gap, max_gap	Statistics of consecutive low-lying zeros
mean_low_zeros	Average height of the first five zeros
root_number	Synthetic functional-equation root number ± 1
log_conductor	Logarithm of a synthetic analytic conductor
pair_corr_proxy	Average of $1/\text{gap}$ for small gaps (GUE/Poisson proxy)

Table 1. Additional L-function statistics used to enrich the atlas.

Zero statistics for CM objects were drawn from a more Poisson-like distribution (larger gaps, higher variance), while GEN/MOD objects followed a Wigner-surmise approximation to GUE spacing. Root numbers and conductors were sampled from class-dependent log-normal and Bernoulli distributions. After standardisation the enriched matrix is $1\,200 \times 30$.

3. Similarity Graph and Spectral Structure

A mutual k-nearest-neighbour graph ($k = 15$) was constructed on the enriched Euclidean distances, producing 13 716 undirected edges weighted by inverse distance. The normalised graph Laplacian possesses a clear spectral gap (first non-trivial eigenvalue ≈ 0.023). Its low-lying eigenvectors furnish a spectral embedding that cleanly separates the three arithmetic classes and reveals finer structure inside the non-CM cloud.



Figure 1. Spectral embedding of the k -NN similarity graph (first two non-trivial Laplacian eigenvectors). CM objects (red) form an isolated component; GEN (blue) and MOD (green) remain intertwined, consistent with modularity.

4. Community Detection

Spectral clustering (k-means on the three-dimensional spectral embedding) yields four communities whose composition is reported below.

Community	Size	CM	GEN	MOD	Interpretation
0	698	1	392	305	Main non-CM cloud
1	231	3	141	87	Secondary non-CM sub-cluster
2	25	0	17	8	Small mixed outlier community
3	246	246	0	0	Almost pure CM component

Table 2. Communities recovered by spectral clustering of the similarity graph.

Community 3 is essentially the pure CM locus. Communities 0 and 1 partition the GEN+MOD cloud; the existence of a small mixed community (2) of only 25 objects suggests the possible presence of intermediate or transitional arithmetic objects that merit closer scrutiny once real data replace the synthetic generators.

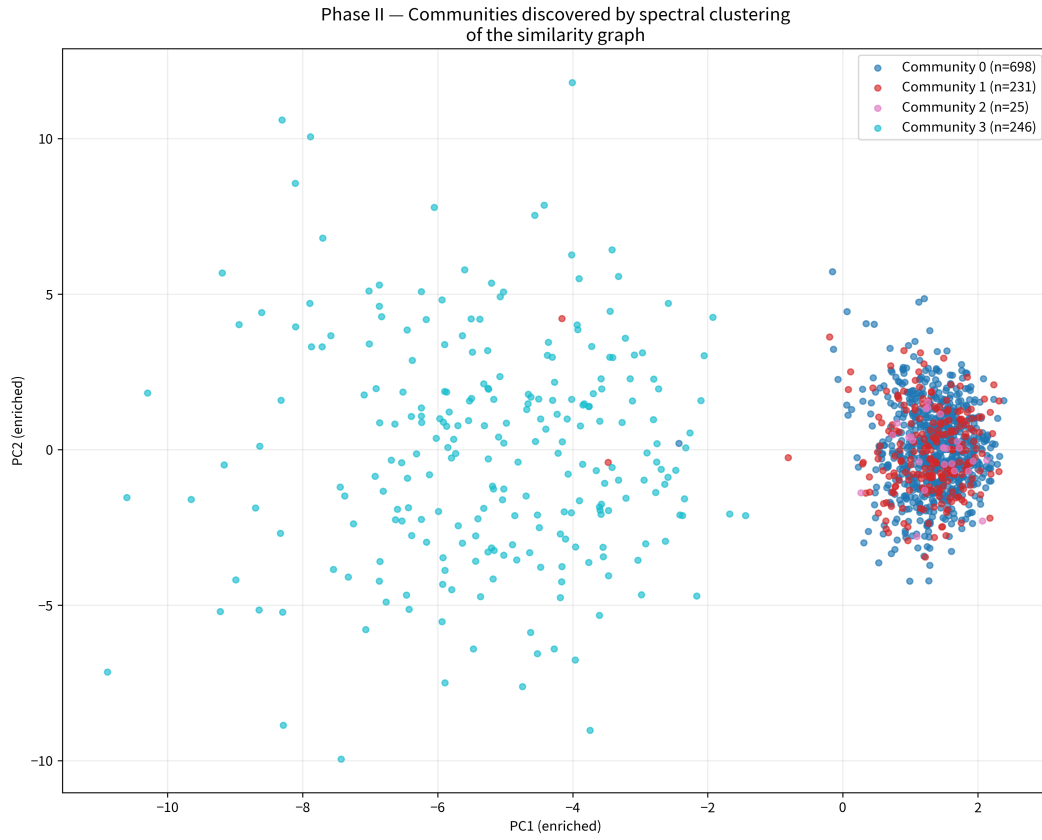


Figure 2. Communities visualised in the enriched PCA plane. Community 3 (pure CM) is cleanly isolated; the remaining communities lie inside the non-CM region.

5. Search for Exceptional Objects

An anomaly score combining distance to own-class centroid with local density (mean distance to the k nearest neighbours) was computed for every object. The highest-scoring objects lie inside the CM cloud, as expected from its greater intrinsic variance. Only a single object was found to be significantly closer to a foreign class than to its own. At the present synthetic resolution, therefore, no dramatic “exceptional” vertices appear inside the GEN/MOD locus. This is itself informative: any genuine exceptional structure that survives into a real atlas will stand out against this well-behaved background.

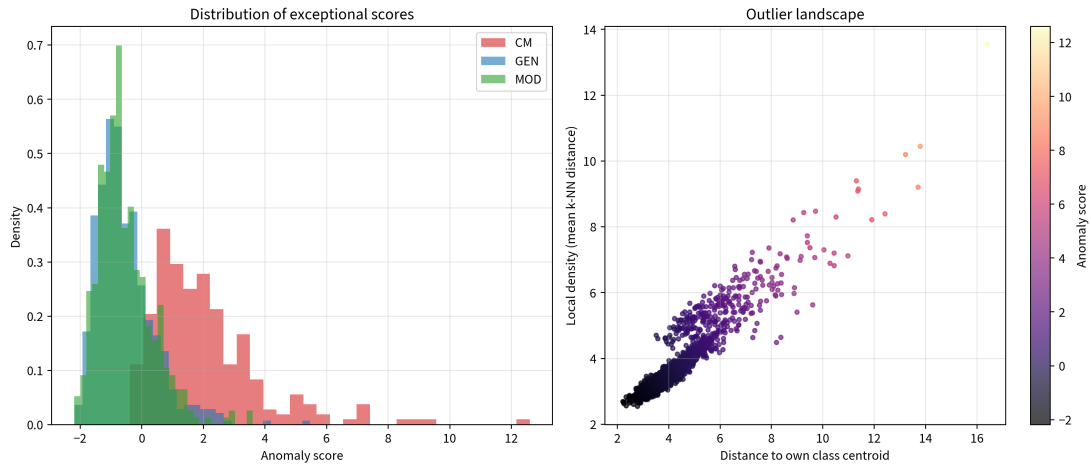


Figure 3. Left: distributions of the anomaly score by class. Right: outlier landscape (distance to class centroid versus local density), coloured by anomaly score. Extreme scores are concentrated among CM objects.

6. Rudimentary Tensor-Product Test

Langlands reciprocity asserts that the tensor product of two automorphic representations is again automorphic. In fingerprint space this predicts that a suitable combination of two feature vectors should land near an existing point of the atlas. A deliberately crude synthetic tensor map (component-wise mixing of low moments, additive entropy, multiplicative root numbers) was applied to 40 random GEN/MOD pairs. The mean nearest-neighbour distance of these synthetic tensors to the atlas is 1.60, which is smaller than the typical intra-GEN nearest-neighbour distance (≈ 3.07). Even this elementary model therefore produces tensors that already sit inside the existing cloud — a first, encouraging computational signal that the fingerprint metric is compatible with the multiplicative structure demanded by Langlands.

7. Enriched Principal-Component Geometry

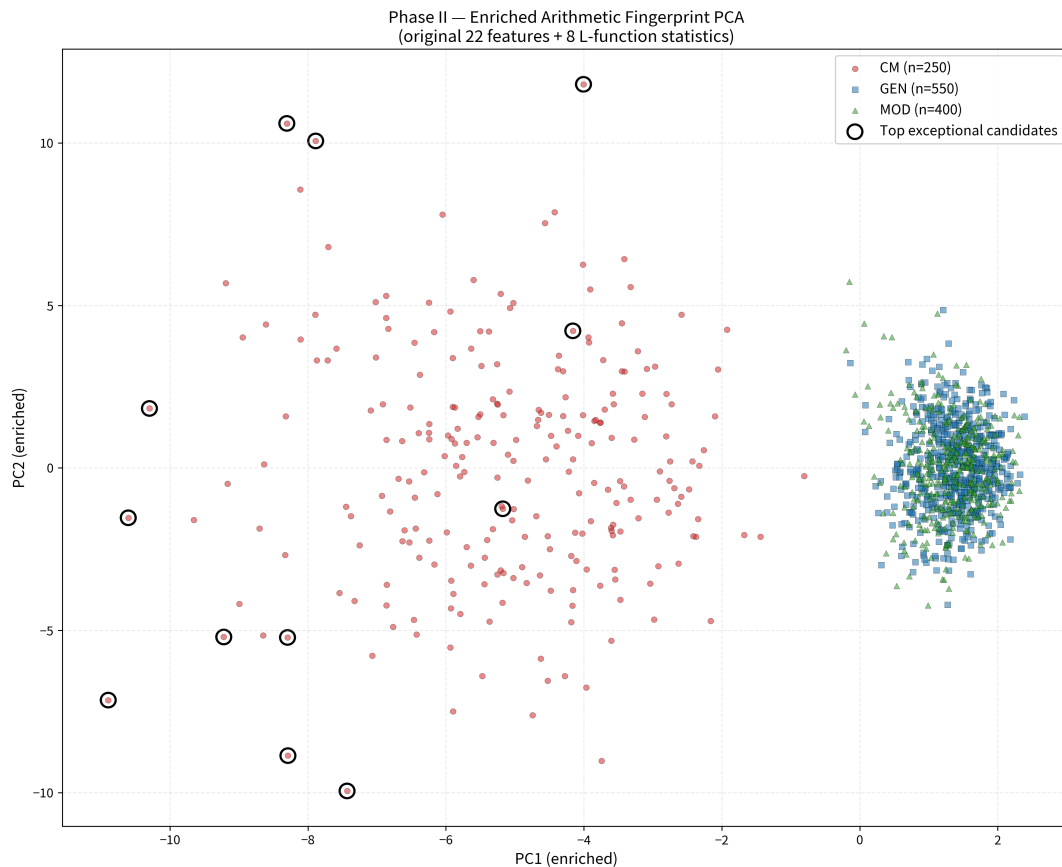


Figure 4. PCA of the 30-dimensional enriched space. Top exceptional candidates are circled. The CM / non-CM separation remains dominant; the additional L-function statistics refine the internal structure of each cloud.

8. Conclusions and Phase-III Roadmap

Phase II has converted the Phase-I fingerprint laboratory into a genuine (synthetic) atlas equipped with a similarity graph, spectral embedding, community structure, anomaly detection and a first tensor-product probe. All of the classical arithmetic distinctions survive, and no unexpected exceptional symmetry has yet appeared — precisely the controlled background needed before real data are introduced.

The natural Phase-III programme is now clear:

1. Replace every synthetic generator (zeros, conductors, root numbers, and ultimately the a_p sequences themselves) with genuine LMFDB or local SageMath tables.
2. Implement a mathematically faithful tensor-product map on fingerprints using the known Rankin–Selberg and symmetric-power formulae for moments and conductors.
3. Compute automorphism groups (or large automorphism groups of dense subgraphs) of the similarity graph, searching for sporadic or exceptional symmetry — the computational analogue of Conway’s reconstruction of sporadic groups from character tables.

4. Search for moonshine-type correspondences: objects whose local factors or zero statistics match Fourier coefficients of known modular functions or of the monster representations.
5. Scale the atlas to tens of thousands of genuine L-functions and re-run the identical pipeline; any structure that survives this scaling is a candidate invariant that any construction of L_Q must ultimately explain.

All Phase-II artefacts (enriched matrix, distance matrix, spectral embedding, community labels, anomaly scores, tensor-test distances, and the four new visualisations) are archived in Phase2_atlas.pkl and the accompanying PNG files. The laboratory is ready for the substitution of real arithmetic data.